

PROFESSIONAL EDUCATION
General Principles and Methods of Teaching

Topics:

- 1. **Instructional Planning**
- 2. **Definition of Terms**
- 3. **Techniques of teaching, discussion, art of questioning, classroom management**

Competencies

- 1. Explain the general principles and methods of teaching
- 2. Distinguish expository and exploratory methods of teaching
- 3. Name the different strategies using the indirect discovery approach
- 4. Use principles of teaching learning associated with instructional operations and curricular management
 - Instructional Planning
 - Stating Objectives
 - Selection of learning activities to achieve objectives
- 5. Select and apply teaching techniques appropriate for particular teaching learning situations.

INSTRUCTIONAL PLANNING: DEFINITION, PRINCIPLE AND COMPONENTS

Instructional Planning

- The ability to visualize into the future—creating, arranging, organizing and designing events in the mind that may occur in the classroom accurate time management and related instances teacher's ability to make decisions about the how and what of teaching

Importance/ Function of Instructional Planning

- Provides an overview of instruction
 - presents a total picture of the lesson for the day or for the year
 - allows some degree of flexibility
- Facilitates good management and instruction
 - provides a classroom script to follow as lessons are conducted
 - shows a clear direction of lessons
- Makes learning purposeful
 - teacher's clear understanding of students' behavior will help increase students' opportunities for learning and reduce anxiety and uncertainty
 - enables the teachers to engage in reflective thinking before making/writing a unit/lesson plan or even during teaching time
- Provides for sequencing and pacing
- Economize time
- Makes learners' success more measurable which assists in reteaching
- Provides for a variety of instructional objectives
- Creates opportunity for higher level questioning
- Assists in ordering supplies
- Guides substitute teachers

Variables in Instructional Planning (Brown, 1988)

- **teacher**- attitudes, beliefs, teacher's content background
- **students**- age, background knowledge, motivational level, interest
- **content**- the type of content that influences the planning process, textbook being used
- **learning content**- subject matter guidelines
- **Material/ resources**- activities and equipment/ tools for teaching are considered first in planning
- **Time**- considerable planning for time

Lesson Plan

- Sets forth the proposed program or instructional activities for each day
- A daily plan
- A step-by-step approach to learning

Components of Lesson Plan

- 1. **objectives**
- 2. **subject matter**
- 3. **learning activities**
 - review/ drills
 - motivation (intrinsic/extrinsic)
 - presentation of the lesson
 - generalization
 - application
- 4. **evaluation**
- 5. **assignment** (for enrichment, extension or remedial)

Instructional Objectives

- Specific and relates to singular subjects and grade levels which includes
 - a. Behavior – Do what?
 - b. Condition – Under what conditions?
 - c. Criteria – How well?
- Aims**
 - General objectives of the Philippine Educational System; broad and value-laden statements about the intent of education
 - Answer the needs and demands of society especially children and youths
 - Formulated by experts
- Goals**
 - Statements that cut across subjects and grade levels
 - Represent the entire school program by professional associations
 - More definite than aims but still non-behavioral and therefore non-observable and non-measurable
- Objective** description of what is to eventually take place at the classroom level that are stated in..
 - Behavioral terms
 - State specific skills
 - Tasks
 - Content attitudes

Domains and Levels of Objectives

Each domain reflects a particular set of beliefs and assumptions about how students learn and behave

- **Cognitive Domain** - objectives that have their purpose to develop students' intellectual skills
 - 1. Bloom's Taxonomy**
 - **knowledge**- emphasis on remembering information
 - **comprehension** – emphasis on understanding and organizing previously learned information
 - **application**- emphasis on using information in pertinent situations
 - **analysis** – emphasis on thinking critically about information by studying its parts
 - **synthesis**- emphasis on original thinking about information by putting its parts into new whole.
 - **evaluation**- emphasis on making judgments about information based on identified standards
 - 2. Anderson's Taxonomy (Revised Bloom's Taxonomy)**
 - Remembering**
 - Understanding**
 - Applying**
 - Analyzing**
 - Evaluating**
 - Creating**
 - 3. Marzano and Kendall's Taxonomy**- consists of three learning system (Cognitive, Metacognitive, and Self-System)
 - Retrieval**
 - Comprehension**
 - Analysis**
 - Knowledge Utilization**
 - Metacognition**
 - Self-System**
- **Affective Domain** - deals with attitudes, values, interest and appreciation which ranges from simple awareness or perception of something to internalizing a phenomenon so that it becomes a part of one's lifestyle
 - **receiving** – emphasis on becoming aware of some communication or phenomenon from the environment
 - **responding** – emphasis on reacting to a communication or phenomenon through participation
 - **valuing** – emphasis on attaching worth to something from the environment evaluating beliefs in the form of acceptance, preference, and commitment
 - **organization** – organizing the values in relation to each other
 - **characterization** – acts in accordance with the accepted value and becomes part of personality
- **Psychomotor Domain** - objectives in this domain are especially appropriate of the objectives generally associated with motor and muscular-skill development
 - **Perception**- Uses the sense organs to obtain cues that guide motor activity; ranges from sensory stimulation (awareness) through cue selection to translation.
 - **Set**- Readiness to take a particular action, includes mental, physical, and emotional set. Perception is an important prerequisite.
 - **Guided response**- Concerned with early stages of learning a complex skill. Includes imitation, trial and error.
 - **Mechanism**- Concerned with habitual responses that can be performed with some confidence and proficiency (less complex).
 - **Complex Overt response**- Skillfully performs acts that require complex movement patterns, like the highly coordinated motor activities. Proficiency indicated by quick, smooth, and accurate performance, requiring minimum effort.
 - **Adaptation**- Concerned with skills so well learned that they are modified to fit special requirements or to meet a problem situation.
 - **Origination**- Creates new movement patterns to fit a particular situation or problems

GENERAL PRINCIPLES & METHODS OF TEACHING

Concepts

- **Approach**- viewpoint toward teaching
- **Strategy**- general design of how the lesson will be delivered
- **Method**- procedure employed to accomplish the lesson objective/s
- **Technique**- style or art of carrying out the steps of a method

Principles Underlying Instruction

- **Principle of Context** – learning depends largely on the setting materials in which the process goes on. This principle comes in different scales of application:
 - textbook only
 - textbook with a supplementary material
 - non-academic and current materials (newspaper, clippings, articles, magazine)
 - multi-sensory aids
 - field experiences; personal, social and community understanding
- **Principle of Focus** – instruction must be organized about a focus or direction
 - Scales of application
Focus established by:
 - page assignment in textbook
 - announced topic together with page or chapter references
 - broad concept or a problem to be solved, or a skill to be acquired to carry on understanding
- **Principle of Socialization** – instruction depends upon the social setting in which it is done.
 - Scales of application
Social patterns characterized by: submission>contribution>cooperation
- **Principle of Individualization** – instruction must progress in terms of the learner's own purposes, aptitudes, abilities and experimental procedures
 - Scales of application
Individualization through:
 - differential performance in uniform tasks
 - homogenous grouping
 - control plan
 - individual instruction
 - large units with optional related activity
 - individual undertakings, stemming from and contributing to the joint undertaking of the group of learners
- **Principle of Sequence** – instruction depends on effective ordering of a series of learning tasks.
 - Sequence is a movement
 - from meaningless emergence of meaning
 - from immediate remote
 - from concrete symbolic
 - from crude discriminating
 - Scales of application
Sequence through
 - logical succession of blocks of content (lesson/courses)
 - knitting learning/lesson/course together by introductions, previews, pretests, reviews
 - organized in terms of readiness
 - organized in terms of lines of emerging meanings
- **Principle of Evaluation** – learning is heightened by a valid and discriminating appraisal of all its aspects
 - Scales of application
 - evaluation of direct results only
 - evaluation related to objectives and process
 - evaluation on total learning process and results

General Approaches to Teaching

Two approaches in teaching: Direct Instruction Approach and Indirect Instruction Approach

Approach	Direct	Indirect
Role	Teacher-centered	Student-centered
	Dispenser of Knowledge, sage on-the-stage, “Ivory Tower”	Facilitator of learning, guide on-the-side
Strategies	Expository	Exploratory
Type of Methods	Deductive	Inductive
	Development	Discovery
	Demonstration	Laboratory
	Lecture	Inquiry/Problem-based/ Problem solving

A Comparison Between Direct and Indirect Approaches	
Direct Approach	Indirect Approach
• makes use of expository strategies • aimed at mastery of knowledge and skills • teacher-oriented • direct transmission of information from teacher • teacher-controlled • highly structured • content-oriented • learner is passive, receives ready information for the teacher	• makes use of exploratory strategies • aimed at generating knowledge for experience • learner-centered • students search for information with teacher's supervision • learner-controlled • flexibly organized • experience-oriented • learner is active in search for information

Direct Instruction Methodology

1. Deductive Teaching/Anticipatory Deduction/Explanatory Deduction	
Concept/s	Steps
- presents a rule/concept/generalization/ principle - illustrate these rule/concept , etc. with examples Anticipatory – forecasts details found in the different students Explanatory – connects facts with principles as interpreted by the teacher	Statement of the Problem - motivation - discussion/relating problem to real life situation
	Generalization solve the problem from given one or two generalization, rules
	Inferring look for the principle that will fit the solution of the problem
	Verification establish validity using references/ materials
2. Showing Method	
Concept/s	Steps
- use for teaching concepts and skills - a teacher-centered strategy that uses teacher explanation and modeling combined with student practice and feedback to teach concepts and skills - widely applicable in different content areas - establishes interaction between teacher and students - helps students to learn procedural knowledge - promotes learning of declarative knowledge - focuses students' attention on specific content/skill - ensures mastery of skills	Introduction review of prior learning
	Presentation - explaining the new concept - modeling the skill
	Guided practice with necessary feedback - provides necessary practice to practice new skills - categorize examples of new concept
	Independent Practice more practice of the skill on concept learned for retention and transfer
3. Lecture/Discussion/Expository Method	
Concept	Steps
- designed to keep students learn organized bodies of knowledge - a teacher-directed model designed to keep learners understand relationship in organized bodies of knowledge - attempts to help students understand not only the concepts but how they are related - based on David Ausubel's concept of meaningful verbal learning - helps learner link new with prior learning and relate the different parts of new learning to each other - designed to overcome the most important weakness of the lecture method by strongly emphasizing learners involvement in the learning process - applicable in different subject areas - ensures clear understanding of information - allows student's participation	I. Planning - Identifying goals - Diagnosing student background - Structuring - Preparing advance organizers
	II. Implementation - Introduction - defining the purpose of the lesson, sharing of objectives and overview to help students see the organization of the lesson - Presentation - defining/explaining major ideas - comprehension monitoring - determining whether or not student understand concepts and ideas - Integration - exploring interconnections between important ideas - Review and closure - summarizing the lecture
4. Demonstration Strategy	
Concept/s	Steps
- a show and tell method - teacher presents and talks about a process, a concept and shows the principles - learner observes and notes down events during teachers' performance	Preparation - motivation - identify objectives/problem/procedure
	II. Explanation of Concepts/Principles/ Process/Theory, etc.
	III. Demonstration of concept process by the teacher, students observe and take down notes
	IV. Discussion of students' observations and answering problems
	V. Verification - justification conclusion

Indirect Instruction Methodologies

Inductive Teaching

Concept/s	Steps
□ It is a discovery learning strategy □ Involves many observable cases or instances that can be compared by the learners □ Students formed own conclusions/ generalization when they are ready to so □ Used when the generalizations are important enough to justify the time spent to the lesson □ Lesson progresses from observations to generalizations	Preparation • apperception • motivation • presentation of the aims of the lesson
	Presentation • teacher/learner cite specific cases of instances which will be the source/ bases of drawing generalization
	Comparison and Abstraction • noting commonalities/differences among the cases, examples cited
	Generalization • Draw generalization for the instances/ examples given from generalization which can be a rule, a formula, principle, concept, etc.
	Applications • Use the newly learned generalizations in new real life situations

Inquiry Approach

Concept/s	Steps
- Engages learners in critical thinking, analysis and problem solving - A systematic and disciplines method of solving and assessing results - Involves testing solution to see if they work and a step-step procedure to solve a problem systematically - Engages students in investigative work - Involves scientific ways of solving problems that include these steps: - Defining problems - Formulation of hypotheses - Gathering data - Analyzing & interpreting data - Making conclusion Forms of Inquiry - Guided Inquiry - Teacher provides data, problems, questions to students - Open inquiry - Students give solutions/ gather data and draw conclusions - Individualized Inquiry - Students work independently	Presentation and clarification of a problem/ issue of inquiry - presentation of objectives - statement of the problem, issue or query - clarification of the problem
	Formulation of Hypotheses - giving tentative solutions to the problem/ query - clarify hypotheses - noting down of hypotheses
	Gathering Data - select references/sources of data/ information - appraise and organize the needed information to answer the problem solving
	Testing Hypothesis
	Formulation of Conclusions Note needs for more study and evaluate procedures
	Application Solving problems using rule/principle etc. in new situations

Laboratory Method

Concept/s	Steps
- Hands-on experiences about materials or facts obtained from research, investigation or experiences - Actual context of learners with the materials/variables - Used to develop skill in basic science process - A discovery experience - Develop centered thinking skills - Involves all students in the manipulative skills - Role of teacher is mostly supervision of activity - Proper handling of tools, equipments, laboratory material (keeping/using/retrieval) - Opportunity to improvise tools/equipment	Preparatory - Motivation - Orientation to the work/activity - Presentation of materials - Precautionary measures
	Supervised Work Period/Laboratory - Students work on the problem (may be the same of different problem) - Teacher supervises the students' work
	III. Culminating Activities - Reporting of group work result/findings - Discussion of the process/findings - Formulation of generalization, rule, concepts, etc. - Application of the discussed concepts/ generalization to new situation

Problem Solving Method	Steps
- An activity that will remove a difficulty or flexibility through reasoning process - May be used; for solving a whole unit/subject as a problem or utilizing problem solving method in a unit of work - Involves scientific ways of searching information - Energizes students to participate activity to find the best solution to a problem - Develop higher order thinking skills (HOTS) - Enhances analytical, logical and create abilities - Involves these steps ✓ Defining problems ✓ Stating/explaining ✓ Critical evaluation of hypothesis/solution ✓ Verification of accepted hypotheses	I. Identification and recognition of the problem
	II. Discussion of key elements of the problem
	Statement of hypotheses Students give/suggests temporary solutions to the problems
	Collection/Gathering Data - Noting down relevant information/ evidences - Interpretation of related evidences - Organizing collected/gathered information
	V. Critical evaluation of suggested solutions or hypotheses
	VI. Verification of accepted hypotheses Testing of given solutions/hypotheses
	VII. Application of the solutions to solve the problem – given problem or another problem

Discovery Teaching

Concept/s	Steps
- Teacher facilitates discovery - Learners participate actively in the learning process - Learning depends on own insights, reflections and experiences (discovery learning) The two discovery teaching strategies are ✓ Inductive discovery ✓ Deductive discovery	<u>Inductive Discovery</u>
	I. Observe and discuss specific examples
	II. Identify and describe common elements of features
	III. Discuss other examples and note common elements
	State main idea based on the common elements against the new examples /elements Check them
	<u>Deductive Discovery</u>
	I. Present an idea that can be verified against evidences
	II. Learners gather/collect finds supporting evidences or examples
	III. Reasoning/Justifying <u>why</u> a certain evidence found is supported to the main idea.
	IV. Students continue searching/finding other evidences to support the given idea.

Project Method

Concept/s	Steps
- A significant practical unit of activity of a problematic nature - Involves planning and carrying out of the planned activities - Students completes certain task in a natural manner - Involves the use of physical materials to complete the unit of experience - Develops sense of cooperation, responsibility to complete a task - Energizes students to evaluate own and other's works (projects) objectively based on developed criteria.	Purposing - Statement of objectives of the projects - Explain the nature of the project - Teacher and students decide on the activities cooperatively
	Executing Student carry out the activities as planned under the guidance of the teacher
	Evaluation - Viewing of finished project - Evaluation by teacher and students based on some decided criteria for the project evaluation

Other Indirect Models/Strategies of Teaching

- **Problem-Based Instruction (PBI)**
 - The essence of PBI consists of presenting students with authentic/meaningful situation that can serve as springboards for investigations and inquiry.
 - This model is highly effective approach for teaching higher-level thinking processes involving:
 - Deriving questions on problem both socially important and personally meaningful to students.
 - Interdisciplinary focus on a particular subject but solutions requires students to deliver into many subjects.
 - Authentic investigation necessitates students to pursue investigation that seek real solution to real problems.
 - Production of artifacts and exhibits requires students to construct products in the form of artifacts and exhibits that represents their solutions.
- **Constructivist Models**
 - Popularized by Piaget and Vygotsky
 - A perspective of teaching and learning in which a learner constructs meaning from experience and interaction with others
 - Teacher provides meaningful/relevant experiences for students from which students construct their own meaning (facilitation)
 - Suggests that learners develop their own understanding of topics they study instead of having it delivered to them by others.
 - Places learner in the center of the learning process why they play an active role in the process of constructing their own understanding.
- **Metacognitive Strategy**
 - Students are trained to become aware of and control their own learning through the metacognitive process.
 - Used when students:
 - plan what strategies to use to meet goal
 - decide what resources are needed
 - monitor own progress
 - evaluate progress
- **Reflective Teaching**
 - Process that enables individual to continually learn from own experiences by considering alternative interpretations of experiences, actions, discussions, beliefs, using introspection and analysis
 - Used when students:
 - Acquire concrete experiences
 - Analyzes experiences
 - From abstractions
 - Apply generalizations to actual situation
- **Cooperative Learning Strategies**
 - Features:
 - Heterogeneously grouped
 - Interdependence among members
 - Individual accountability
 - Explicit teaching of collaborative skills
 - Variants:
JIGSAW
 - How/when used:
 - A group are formed
 - Material is divided into sections
 - One member takes care of a section of the material
 - Each member meets with those from other groups who are assigned to a similar section
 - Members discuss/work on the material
 - Return to their previous group to inform others in their group
 - Testing students after the -puzzle is completed
- **Student Teams Achievement Strategy (STAS)**
 - A type of group work activity in which students interact together to master a specific academic material
 - How/When Used:
 - Information is presented;
 - Students are divided into learning teams to master lessons using worksheets;
 - Discussion, tutoring, quizzing one another;
 - Scores from tests are recorded; and
 - There is improvement from the previous achievement score of the team, additional points are given.

Some Essential Teaching Techniques

- **Discussion**
 - An attempt to get away from the traditional classroom procedure of question and answer and recitation style; it is also a tool for implementing the democratic process in teaching and learning transaction.
 - To get the desired understanding through analysis and evaluation of facts by encouraging group thinking as learners/participants attempt to find solutions to a problem

Types of Discussion Procedures

Panel- Forum-A direct, conversational, interactional discussion among a small group of experts or well-informed lay persons.

- Like any intelligent conversation, except that participant speak loud enough for the audience to hear
- Success depends on the preparations done before the meeting, the wise selection of participants, and the discussant leader
- *Guidelines to Ensure a Successful Panel-Forum*
 - The leader or chairperson
 - Emphasizes tactfully the purpose and philosophy of the discussion;
 - Keeps the conversation moving from point to point;
 - Makes sure that each panel member has a chance to express his/her views, to ask questions intended to clarify points for the audience.
 - Maintains an impartial position by refraining from participating as a member of the panel.
 - Must be group-oriented rather than self-centered; and
 - Summarizes discussion briefly and invites comments or questions from the audience
 - The participants in the panel
 - Talk in a formal manner; and
 - Contribute to the method of public conversation meeting by giving brief remarks to each other but should be heard by the audience

Symposium-Forum

- More formal than the panel discussion
- Persons with special competence deliver uninterrupted speeches on different aspects of a problem, and these are followed by a forum period
- Essentially a public-speaking program; not conversational
- *Guidelines to Ensure a Successful Symposium Forum*
 - Organizing a symposium includes:
 - Deciding the purpose of the meeting
 - Choosing and framing the topics to arouse interest
 - Choosing speakers (the number of speakers depends on the number of significant sources of information or points of view that should be considered)
 - Choosing a chairman
 - Briefing the chairman and the speakers on the objective of the symposium and in the procedures to be followed
 - Speakers should not forget that discussions mean -thought in process, ll that its purpose is to help listeners analyze the problem and not to make conclusions for them
 - In the discussion process, the chairman or the moderator should help bring together the thoughts of the speakers as the program unfolds. The chairperson also makes a proper transition to the forum period.
 - The chairperson or moderator sees to it that the important questions about he issues presented are not neglected before the adjournment of the forum.

Debate

- A discussion that occurs when people with different beliefs study the same problem and arrive at different conclusions;
- A more formal type in which each participant makes a prepared speech for or against a proposition;
- Debaters are usually allotted equal time to speak to present an analysis of a problem and a fair presentation of the arguments for or against it.
- ***Guidelines to Ensure a Successful Debate***
 - Attitude in debate is properly adjusted in relation to the subject, the opponents, and the listening audience. Arguments are directed toward the person. A proper attitude includes keeping one's perspective, temper and sense of humor.
 - Refutation, or the answering of opponents' arguments, should be woven throughout the main speeches in a debate. There are a number of ways of answering an argument that is to be refuted; including
 - Pointing out that it is not relevant or important to the question
 - Showing that it is not supported by the facts or that insufficient evidence has been given
 - Indicating fallacies in reasoning
 - Arriving at a contrary argument by sound reasoning
 - Supplying more and better evidence to support one's side of the argument
 - Turning the argument so that it actually helps one's side
 - Make each point consistent with those made by colleagues who are also upholding a similar view.
 - In answering point of the other side, it is well to have a large supply of evidence that can be drawn from to support one's position.
 - A knowledge of the audience on the question will be helpful in determining whether the audience is favorably or unfavorably disposed to what is being said.
 - A card-index file of all points and evidence that may be used should be kept; with some system worked out, so that the members of the team can quickly draw on the material they want to include at any given stage in the debate.

Round Table Conference

- A small discussion group seated face to face around a table, without a larger audience; a small conference by another name and has the nature of an informal semi social gathering
- Members will not be hearing speeches but do their own talking

Guidelines to Insure a Successful Round Table Conference

- All members should have a precise understanding of what is to be delivered
- A leader whom the group likes and respects is chosen to lead and prepare for the meeting
- Procedure in conducting the session of a round table conference is summarized below:
 - Introducing remarks, stating the question to be discussed in as interesting a manner as possible
 - Statements of the facts or a brief story of a real or fictitious case
 - Presentation of agenda
 - Group discussion of each of the issues in the agenda
 - Summary of the discussion
 - Consideration of what action to take as a result of the discussion

Special Techniques to insure active participation in the forum period:

- a. Role playing*
 - Spontaneous acting out of problems or situations
 - Portray a situation more candidly
- b. Case Study*
 - Another group centered procedure which presents specific situations or problems to stimulate discussion
 - This technique implies extensive analysis and interpretation of a case selected to demonstrate a learning outcome
- c. Buzz Session*
 - Is used when dealing with familiar topics that need group opinion, evaluating, planning or interaction
 - Involves groups not exceeding six persons
 - A leader and a secretary are chosen to lead the discussion and to record what have been discussed
 - Each group is given the time to present the questions or outcomes of their discussion
- d. Workshop*
 - Involves the use of group process in attacking and solving problems
 - Persons with problems of common concern come together to attack and solve their problems cooperatively
 - Makes use of a variety of means and devices in the solution of problems such as group meeting, individual conference, field trips, excursions and the use of resource persons and consultants
 - Values gained from the workshop are both intellectual and social
- e. Seminar*
 - Deliberately looks for the solution to the problems from the evidence based on reading, experiences and minds of the participants
 - Attempts to develop a policy or solution that is better than what is in existence

ART OF QUESTIONING

What is Questioning? Key technique in teaching

- **Purposes of Questions**
 - Arouse interest and curiosity
 - Review content already learned
 - Stimulate learners to ask questions
 - Promote thought and the understanding of ideas
 - Change the mood/tempo, direction of the discussion
 - Encourage reflection and self evaluation
 - Allow expressions of feelings

Types of Questions

- According to thinking process involved:
 - Low-level questions- focused on facts, don't test level of understanding or problem solving skills
 - Examples: Who declared martial law? What important events happened in WW II?
 - High-level questions- go beyond memory and factual information, more advance, stimulating and more challenging, involve abstraction and point of view.
 - Examples:
 - How did the recent war between the government forces and MILF affect the people in Mindanao? What alternative could we practice to become attentive?
- According to type of answer required:
 - Convergent questions- tend to have on correct and best answer.
 - Are used to drill learners on vocabulary, spelling and oral skills but not appropriate for eliciting thoughtful responses
 - Usually start with what, who, when, or where
 - Are referred to as low-level questions
 - Are useful when applying inductive approach and requires short and specific information from the learners
 - Divergent questions- open-ended and usually have many appropriate answers.
 - Reasoning is supported by evidence and examples
 - Associated with high level thinking processes and encourage creative thinking and discovery learning

- According to the cognitive taxonomy
 - 1st Level: Knowledge- memorize, recall, label, specify, define, list, cite, etc
 - 2nd Level: Comprehension- describe, discuss, explain, summarize translate, etc
 - 3rd Level: Application- solve, employ, demonstrate, operate experiment, etc.
 - 4th Level: Analysis- interpret, differentiate, compare invent, develop, generalize
 - 5th Level: Synthesis: Invent, develop, generalize
 - 6th Level: Evaluation- Criticize, judge, interpret
- According to the questions used by teachers during open discussion
 - **Eliciting Questions**- these are employed to : encourage an initial response; encourage more students to participate in the discussion; rekindle a discussion that is lagging or dying out
 - **Probing Question**—seek to: expand or extend ideas; justify ideas; clarify ideas
 - **Closure-seeking Questions**- used to : help students form conclusions, solutions or plans for investigating problems.
 - Referential questions** – are questions that seek to draw response from students which a teacher has no expected answer
 - Display Questions**- are questions in which a teacher expects a correct answer from the student

▪ Guidelines in Asking Questions

- Wait Time- the interval between asking a question and the student response.
- Prompting- uses hints and techniques to assist students to come up with a response successfully
- Redirection- involves asking of a single question for which there are several answers; used in a high level questioning.
- Probing- a qualitative technique used for the promotion of effective thought and critical thinking; provides the students a chance to support or defend a stand or point of view
- Commenting and prompting- used to increase achievement and motivation

▪ Tips on Asking Question

- Ask questions that are:
 - stimulating/thought-proving
 - within student's level of abilities
 - relevant to students daily life situation
 - sequential- a stepping stone to the next
 - clear and easily understood
- vary the length and difficulty of questions, phrase questions clearly
- have sufficient time for deliberation
- follow up incorrect answer
- call on volunteers or non-volunteers
- call on disruptive students
- move around the room for rapport/socialization
- encourage active participation

Determining Appropriate Evaluation Instruments

The concept of evaluation

- aims basically to determine student mastery of what has been taught.
- is a two-part process:
 - measuring pupils' individual performance
 - judging about the adequacy of the – may use one of two major approaches in determining how well a pupil has performed:
 - *norm-reference assessment*, where a pupils' performance is compared to the average performance of his/her classmates;
 - *criterion – reference assessment*, where the rating is based on comparison of a students' performance with a pre-determined standard instruments used for evaluation may be:
 - *informal evaluation* – depends on teacher's observations of a variety of pupil performances as they do learning tasks, complete projects, or interact with others; requires teachers to make judicious inferences about what learners can and cannot do; may use this question as a basic guide in selecting informal assessment techniques, -Will the procedure provide the information/ need to make an adequate judgment about a child's performance?||
 - *formal evaluation* – includes teacher-prepared tests and commercially-available standardized tests, such as rating scales, learning checklists, essay tests, true-false test, multiple-choice tests, completion tests, matching tests, etc.
 - may also be used to:
 - assess progress of individual learners
 - evaluate own performance of teacher
 - refine instructional plans and/or provide instruction to clear up understandings of certain topics taught
 - provide basis for instructional planning when the same content tested is taught again
 - has a need for good record keeping to monitor the
 - progress/development of learners overtime

When to use some test types.

- use informal evaluation tools when looking for specific behaviors that learners are expected to demonstrate
- use rating scales and checklist when judgment about several levels of performance quality is needed
- use essay tests especially for upper grade levels when determining students to put together isolated process of information in a meaningful way
- use true-false test when covering a broad range of content
- use matching test if pupils' grasp of association is to be tested
- use completion test to sample only a cross-section of content and to eliminate guessing among pupils

What evaluation type to use during the instructional act

a. Prior to Instruction: Pre assessment

- done to determine the learner's entry behavior (what knowledge/skills/attitudes they already know or still need to know) before objectives are set or before instruction begins for maximum learning purposes
- involves use of such instruments as readiness test, aptitude test, pre test on course objectives, or observational techniques

b. During Instruction: Formative Evaluation

- provides on-going feedback to the teachers and students regarding their success or failure during instruction;
- helpful in deciding whether changes in subsequent learning experiences are needed
- helpful in determining specific learning errors that need correction
- usually makes use of paper and pencil tests and observational methods

c. After Instruction: Summative Evaluation

- is provided to determine >how well students have learned/attained instructional objectives>how well instruction was done

CLASSROOM MANAGEMENT

What is Classroom Management?Refers to the operation and control of classroom activities; involves the ability to maintain order and sustain pupil attention

What are the Purposes of Classroom Management?

- To minimize the occurrences of discipline problems
- To increase the proportion of classroom time devoted to constructive and productive activity.

3 C's of Classroom Control

- Content- facilitate the delivery of instruction
- Conduct- promotion of orderly and safe learning environment
- Context- emphasis is on communication rather than physical elements associated with classroom setting

Types of Control

- Preventive Control- aimed at minimizing the onset of anticipated discipline problems through planning
- Supportive Control- aimed at directing student' behavior before it becomes a full blown problem
- Corrective Control- seeks discipline student' behavior before it becomes a full standard of good conduct

Nature and Dynamics of Approaches to Classroom

Management Behavior-Modification Approach

- Based on principles of behavioral psychology: -All behavior is learned (Sulzer and Mayer)
- Built on two assumptions:
 - Learning is controlled largely, if not entirely, by events in the environment.
 - There are four processes that account for learning at all age levels and under all conditions (positive reinforcement, negative reinforcement, extinction or time out and punishment)

A behavior is shaped by consequence (what consequence follows a behavior).

Some situations that illustrate the use of Behavior Modification Approach

- Ben prepares a neatly written paper, which he submits to the teacher. The teacher praises Ben's work and comments that neatly written papers are more easily read than those which are sloppy. In subsequent papers, Ben takes great
- Susan whose neat work has always been praised by the teacher, prepares a neatly written paper, which she submits to the teacher. The teacher accepts and subsequently returns the paper without comment. Susan becomes less neat in subsequent papers. >Extinction is the withholding of an anticipated reward in an instance where that behavior was
- The students in Ms. Tan's English class have come to expect that she will give them the opportunity to play a word game if their work is satisfactory. This is the activity they will enjoy. Miss Tan notes that all their papers were neatly done except Jim's paper. She tells Jim that he will not be allowed to participate in the class game and must instead,
- Jim prepares a rather sloppily written paper, which he submits to the teacher. The teacher rebukes Jim for failing to be neat, informs him that sloppily written papers are difficult to read, and tells him to rewrite and resubmit the
- Jim is one student in the class who consistently presents the teacher with sloppy papers. Despite the teacher's constant nagging of Jim, his work becomes no neater. For no apparent reason, Jim submits a rather neat paper. Miss Tan accepts it without comment-and without the usual nagging. Subsequently, Jim's work becomes neater. >Negative

Guidelines for Using Punishments

- Don't threaten the impossible. Make sure the punishment can be carried out.
- Don't punish when you are at loss of what else to do in an emotional state. The quiet cool approach is more effective than the angry, emotional approach.
- Don't assign extra homework as well as the subject
- Be sure the punishment follows the offense as soon as possible. Don't impose the punishment two days after the student misbehaves
- Be sure the punishment follows the offense as soon as possible. Don't overact mild behavior or underplay or ignore serious misbehavior.
- Be consistent with punishment. If you punish one student for something, don't ignore it when another student does the same thing.
However, students and circumstances differ, and there should be room for modification
- Don't use double standards when punishing. You should treat both sexes the same way, and low-achieving students the same way.
- Give the students the benefit of doubt. Before accusing or punishing someone, make sure you have the facts right
- Don't hold grudges. Once you punish the student, put the incident behind and try to start with a clean slate.
- Don't personalize the situation. React to misbehavior, not to the student
- Document all serious incidents. This is especially important if the behavior involves sending the student out of the room or possible suspension.

Socioemotional-Climate Approach

- Has its roots in counseling and clinical psychology (Carl Rogers)
- Places great importance on interpersonal relationships
- The teacher is the major determiner of interpersonal relationships and classroom climate
- Attitudes that are essential in effective facilitation of learning (Rogers)
- Realness, genuineness and congruence – realness is the expression of the teacher being himself or herself, the teacher is aware of his/her feelings, accepts and acts on them and is able to communicate them when appropriate... allows the teacher to be perceived by students as a real person
- Acceptance, prizing, caring and trust – behaviors that makes students feel trusted and accepted
- Emphatic understanding- using student's point of view, sensitive awareness of the student's feelings and is nonevaluative and nonjudgmental

Principle of Communication (Ginott, Teacher and Child)

- The teacher talks to the situation and not to the personality and character of the child
- The teacher shows the ability to describe what she/he saw, describe how he/she feels and describe what needs to be done.

Glasser' View: The importance of Teacher Involvement (Schools Without Failure)

- Student misbehavior is the result of the student's failure to develop success identity
- Suggests that a teacher should:
 - Become personally involved with the student; accept the student but not the student misbehavior
 - Elicit description of the student's present behavior
 - Assist student in making a value judgment about the problem behavior
 - Help student plan a better course of action
 - Guide student in making a commitment to the course of action he/she selected
 - Reinforce the student as she follows the plan and keeps the commitment
 - Accept no excuses if the student fails to follow through with her commitment
 - Allow the student to suffer the natural and realistic consequences of misbehavior

Use of logical consequence

- LC express the reality of the social order; LC results from a violation of an accepted social rule (Punishment expresses the power of a personal authority)
- LC are logically related to the misbehavior; the student sees the relationship between the misbehavior and its consequence
- LC involve no element of moral judgment; student's misbehavior is viewed as a mistake, not a sin
- LC is concerned with what will happen next; the focus is on the future (Punishment is in the past)
- LC are involved in a friendly manner (Punishment involves either open or concealed anger); the teacher should try to disengage himself from the consequence

Group –Process Approach

- Also known as sociopsychological approach – based on the principles from social psychology and group dynamics (Schmuck, Johnson and Bany and Kounin)
- Based on the following assumptions:
 - Schooling takes place within a group context- the classroom group
 - Teacher's task is to establish and maintain an effective, productive classroom group
 - The effective, productive classroom group is characterized by certain conditions that are compatible with the properties
 - of a social system
 - The classroom management task of the teacher is to establish and maintain such conditions

Six properties of Classroom Management (Schmuck)

- Leadership – creating a climate in which students perform leadership functions
- Attraction – refers to the friendship patterns in the classroom group
- Norms – shared expectations of how group members should think, feel and behave
- Communication – the vehicle through which the meaningful interaction of members takes place and through group processes in the classroom occur
- Cohesiveness – collective feeling that the class members have about in the classroom group
- Facilitation and Maintenance (Johnson and Bany)
 - Facilitation – refers to management behaviors that improve conditions within the classroom
 - Maintenance – management behaviors that restore or maintain effective condition

Four Kinds of Facilitation Behaviors

- Achieving unity and cooperation
- Establishing standards and coordinating work procedures
- Using problem solving to improve conditions
- Changing established patterns of group behavior

Three Kinds of Maintenance Behaviors

- Maintaining and restoring morale
- Handling conflict
- Minimizing management problems

Kounin	Classroom Management
Withitness	“eyes at the back of the head”, teacher knows what’s happening inside the classroom
Momentum	Continuous lesson
Overlapping	Multi-tasking, can do 2 or more things at the same time
Flip-flop	Jumping from one topic to another
Truncation	Ending abruptly the lesson
Dangling/Dangle	Lesson is left hanging
Thrust	Lesson without student’s readiness

	Classroom Management
Stimulus-bounded	Easily distracted
Hurdle Help	Teacher helps the student
Proximity Control	Teacher goes near the noisy student
Signal interference	“Shhhh!”, Non verbal cues
Antiseptic bouncing	“Get out!”
Direct Appeal	Teacher confronts the student
Planned Ignoring	Teacher ignores the noisy student

Techniques of Building Good Discipline:

- Demonstration. Students know exactly what is expected. In addition to having expected behavior explained to them, they see and hear it.
- Attention. Students focus their attention on what is being depicted or explained. The degree of attention correlates with the characteristic of the model (teacher) and characteristics of students
- Practice. Students are given opportunities to practice the appropriate behavior.
- Corrective feedback. Students receive frequent, specific, and immediate behavior is suppressed and corrected.
- Application. Students are able to apply their learning in classroom activities (role playing, modeling activities) and other real-life situations

Strategies for Managing Students with Problems

- Accept students as they are, but build on and accentuate their positive qualities
- Be yourself. Students can recognize phoniness and take offense at such deceit.
- Be confident. Take charge of the situation, and don't give up in front of the students.
- Provide structure. Many of these students lack inner control and are restless and impulsive.
- Explain your rules and routines so students understand them. Be sure your explanations are brief; otherwise you lose your effectiveness and you appear to be defensive of preaching
- Communicative positive expectations that you expect the students to learn and you require work.
- Rely on motivation, and not on your prowess to maintain order; an interesting lesson can keep the students on task
- Be a firm friend, but maintain psychological and physical distance so your students know that you are still the teacher
- Keep calm, and keep your students calm, especially when conditions become tense or upsetting. It may be necessary to delay action until after class when emotions have been reduced
- size up the situation and be aware of undercurrents of behavior, since these students are sizing you up and are knowing manipulators of their environment
- Anticipate behavior, being able to judge what will happen if you or a student decide on a course of action may allow you to curtail many problems.

Establishing Routines

- Beginning and ending the day or class period
- Transitions
- Getting/distribution of materials and equipment
- Group work
- Seatwork and teacher-led activities

Effective Questioning and Reacting Techniques

1. Varying type of question
2. Asking non-directed questions
3. Calling on non-volunteers
4. Rephrasing
5. Sequencing logically
6. Requiring abstract thinking
7. Asking open-ended questions
8. Allowing for sufficient wait time

Functions of Homework

1. Pre-learning
2. Checking for understanding
3. Practice
4. Processing

Principles in the Use of Homework

1. The purpose of homework and outcome of homework should be identified and articulated.
2. Homework tasks should be differentiated by:
 - a. Difficulty or amount of work
 - b. By the amount of scaffolding provided; and
 - c. Learner's learning style or interest.
3. Parent involvement in homework should be kept to a minimum.
4. If homework is assigned, it should be commented on.
5. Move from grading to checking.
6. Establish and communicate a homework policy.
7. Establish a homework support programs.

Building and Enhancing New Literacies Across the Curriculum

21st Century Education

1. Focus on a project-based curriculum
2. There is an innovation in education
3. Schools will go from 'buildings' to 'nerve centers'
4. Teachers from dispensers of knowledge to facilitators of learning

Attributes of 21st Century Education

1. Integrated and Interdisciplinary
2. Technology and Multimedia
3. Global Classrooms
4. Creating to Constant Personal and Social Change and Lifelong Learning
5. Student-Centered
6. 21st Century Skills
7. Project-based and Research Driven
8. Relevant, Rigorous, and Real World

Characteristics of 21st Century Teachers

1. Multi-literate
2. Multi-specialist
3. Multi-skilled
4. Self-directed
5. Lifelong learner
6. Flexible
7. Creative problem solver
8. Critical Thinker
9. Has passion
10. High EQ

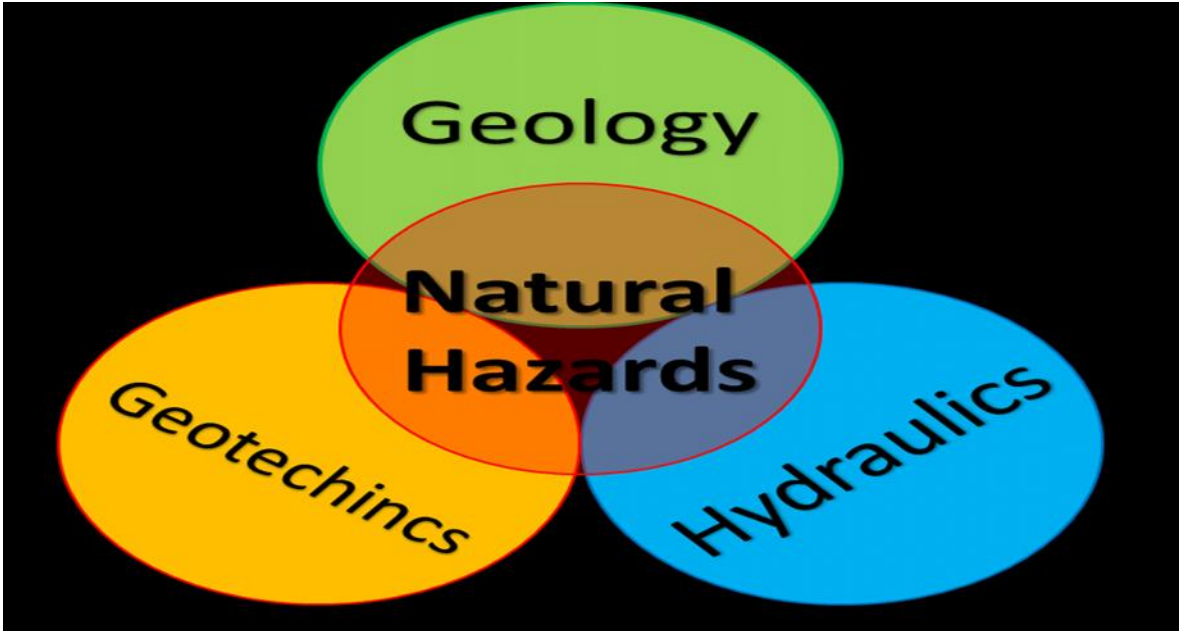
Technology Tools of 21st Century Education

1. Affinity Groups
2. Blogs
3. E-portfolio
4. Hypertext
5. Podcasts
6. Web 2.0
7. Myspace
8. Second Life
9. Semantic Web
10. Webkinz

11. Wiki
12. YouTube
13. Google
14. Prezi
15. Easybib
16. Social Media Platforms
17. Smartboards and Audience Response Systems
18. ReadWriteThink.org
19. WebQuest Page
20. Literacy Web

Approach to Integration

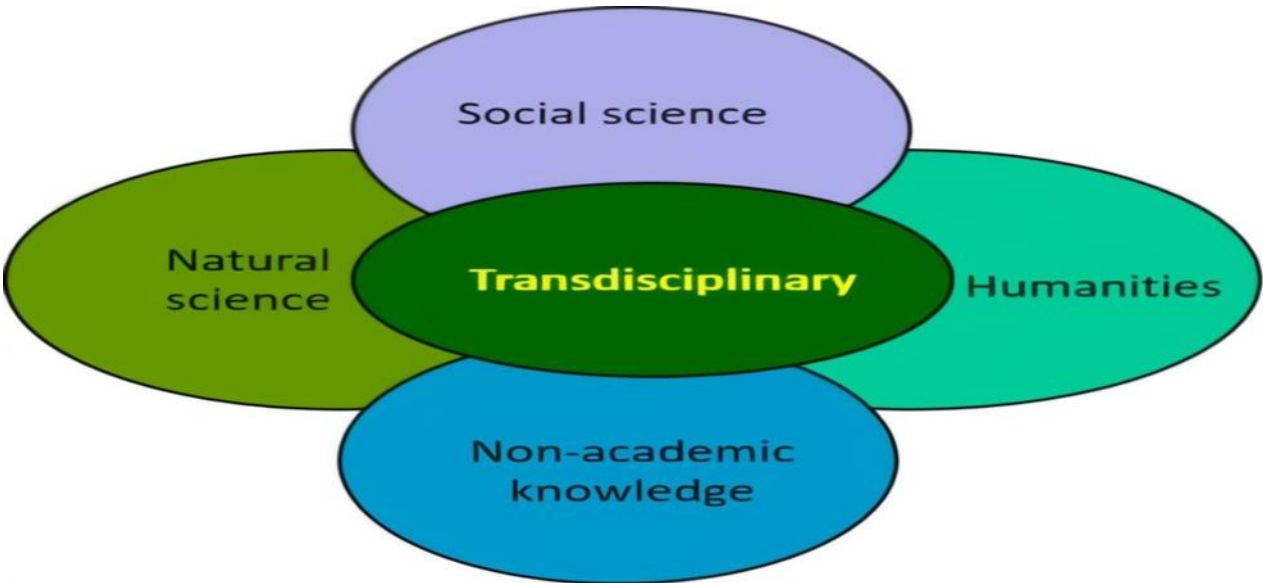
1. Multidisciplinary Approach



2. Interdisciplinary Approach



3. Transdisciplinary Approach



Methods of Curriculum Integration

1. Project-Based Learning
2. Service Learning
3. Learning Centers/ Parallel Disciplines
4. Theme-based
5. Fusion

Multicultural Literacy – consists of the skills and ability to identify the creators of knowledge and interests to reveal the assumption of knowledge from diverse ethnic and cultural perspective.

Global Literacy – aims to address issues of globalization, racism, diversity, and social justice.

- It requires awareness and action, consistent with a broad understanding of humanity, the planet, and the impact of human decision on both.

Dimensions of Global Education

Dimension 1: Examine issues local, global, and cultural significance.

Dimension 2: Understand and Appreciate the perspective world views or others

Dimension 3: Engage in Open, Appropriate, and Effective Interactions across Cultures

Dimension 4: Take action for collective well-being and Sustainable Development

Social Literacy – entails the development of social skills, knowledge, and positive human values toward desire and ability in human beings to act and react positively and responsibly in a wide range of complex social settings.

Types of Social Skills

1. Effective Communication
2. Conflict resolution
3. Active listening
4. Empathy
5. Relationship management
6. Respect
7. Problem-solving skills
8. Interpersonal skills

Media Literacy – the ability to access, analyze, evaluate, and create media.

- The ability to identify different types of media from wide array of sources and understand the messages they bring.

Roles of Media Literacy

1. Learn to think critically
2. Become a smart consumer of products and information
3. Recognize point of view
4. Create media responsibly
5. Identify the role of media in our culture

Media and Information Literacy – a combination of knowledge, skills, and practices required to access, analyze, evaluate, and use information and knowledge in creative, legal, and ethical ways.

Dimension of Media and Information Literacy

1. Tool Literacy
2. Resource Literacy
3. Social-Structural Literacy
4. Research Literacy
5. Publishing Literacy
6. Emerging Technology Literacy
7. Critical Literacy

Integrating Media Literacy in the Curriculum

1. Teach students to evaluate media.
2. Show students where to find digital resources and databases.
3. Compare/Contrast various media sources
4. Discuss how media edits and alters
5. Examine the truth in advertisement
6. Have students create media

Financial Literacy - is a core life skill in an increasingly complex world where people need to take charge of their own finance, budget, financial choices, managing risks, saving, credit, and financial transactions.

Financial Plan Steps:

1. Calculating the net worth
2. Determining the cash flow
3. Considering the priorities

Spending Strategies:

1. Start by listing your goals
2. Divide your goals accordingly
3. Estimate the cost
4. Project future cost

5. Calculate how much you need to set aside
6. Prioritize your goals
7. Create a schedule for meeting your goals

Investment and Investing Aspects

1. How long will you invest the money? (Time Horizon)
2. How much money do you expect your investment to earn? (Expectation of Investment)
3. How much of your investment are you willing to lose in the short-term in order to earn more in the long term? (Risk Tolerance)
4. What types of investment interest you? (Investment type)

Emergency Savings Fund

1. To become financially independent
2. To save on everything you buy
3. To prepare for the future
4. To get out of debt
5. To respond to emergencies

Common Financial Scams

1. Phishing
2. Social media scams
3. Phone scams
4. Stolen Credit Card Numbers
5. Identity Theft

Things to Avoid

1. Never wire money to a stranger
2. Don't give out financial information
3. Never click on hyperlinks in emails
4. Use difficult passwords
5. Don't download software from pop-up windows

Digital Literacy - Is the set of competencies required for full participation in society that includes knowledge, skills, and behaviors involving the use of effective use of digital devices.

Cybersecurity - is the practice of protecting systems, networks, and programs from digital attacks.

Cyber Citizenship - is an old value that is now applies to a new medium, the Internet. It means being responsible when using the internet.

Eco-Literacy - Considers ecological systems and awareness of how society operates within natural aspects as an educational imperative.

Practices

1. Develop empathy for all forms of life
2. Embrace sustainability for all forms of life
3. Make the invisible visible
4. Anticipate unintended consequences
5. Understand how nature sustains life

Integrating Ecological Literacy into the Curriculum

1. Place-Based Learning
2. Project-Based Learning
3. Socratic Inquiry
4. Experiential Learning
5. Interdisciplinary Learning

Arts and Creative Literacy - It's the act of turning new and imaginative ideas into reality.

Creativity - is the process of having original ideas that have value.

Seven Habits of Highly Creative People

1. Prepare the ground
2. Plant seeds for creativity
3. Live in the questions
4. Feed your brain
5. Experiment and explore
6. Replenish your creative stock
7. Liberate your creativity

Eye-Hand Coordination - Coordinated control of eye movement with hand movement and the processing of visual input to guide reaching and grasping along with the use of hands to guide the eyes.

Visual Literacy - Refers to interpreting and creating visual images

Verbal Creativity

1. Fluency
2. Flexibility
3. Originality
4. Elaboration

Aesthetics - Aesthetic or Esthetics is the philosophical study of beauty and taste.

Approaches:

1. It is the study of aesthetic concepts
2. It is the study of mind, emotions, attitudes, etc.
3. It is the study of the aesthetic object

Integrating Arts and Creativity Literacy

1. Physical environment
2. Emotional environment
3. Project-based Learning
4. Teach Creative thinking skills
5. Alternative assessments
6. Scheduling
7. Student Centered and Personalized Learning
8. Incorporate arts
9. Integration of technologies
10. Preparing the body and brain for creativity